AI COURSE MINI-PROJECT

AI-Based Smart Waste Segregation and Classification System

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CONTEXT

Introduction and Motivation

THE PROBLEM

A growing waste crisis

- Global municipal waste is a mounting crisis for cities worldwide.
- Landfills overflow, and decomposing waste releases methane (CH4).
- Manual segregation at the source fails: it is inconsistent, slow and unhygienic.

THE AI ANSWER

Why AI is relevant

- Computer vision classifies waste automatically from a single image.
- Real-time results replace slow, unhygienic human sorting.
- Lightweight models run in a web browser or on edge devices.

GOALS

Project Objectives

01

Waste classification model

Classify waste images into 5 categories: Plastic, Paper, Metal, Glass and Organic Waste.

02

MobileNetV2 with transfer learning

Reuse pre-trained ImageNet features to build a lightweight, accurate classifier.

03

Interactive web application

Support file upload and live webcam capture for instant predictions.

04

Confidence and bin guidance

Display model confidence with a bin recommendation: **Blue**, **Green** or **Yellow**.

RELATED WORK

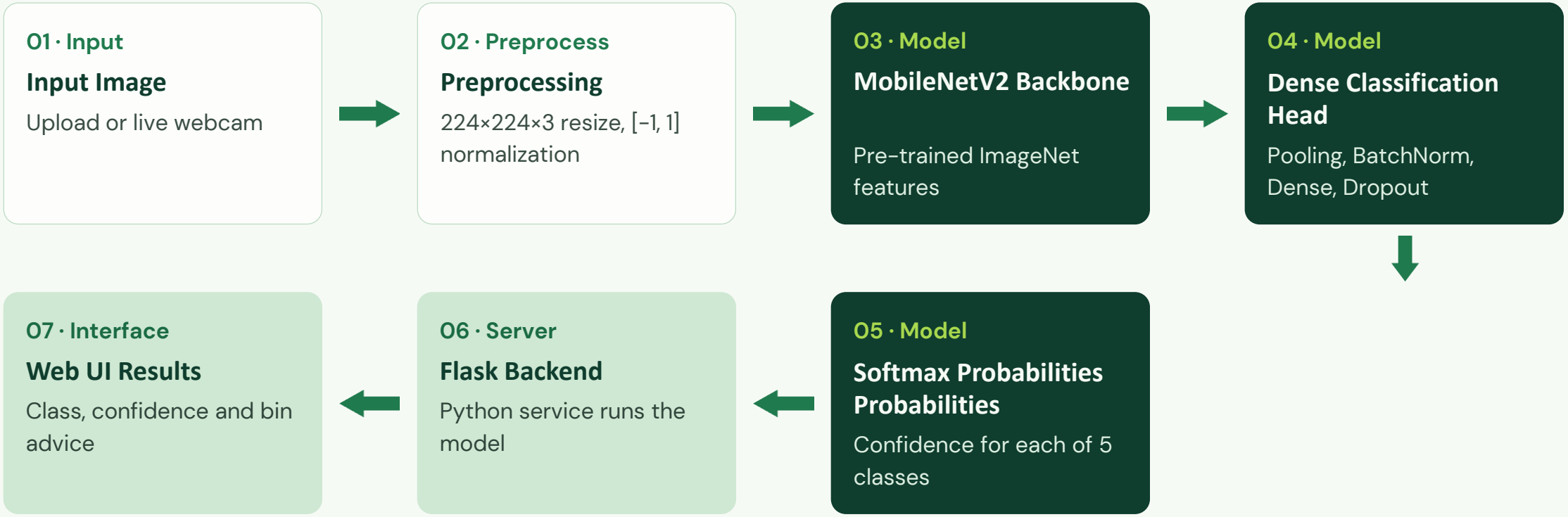
Literature Review

#	Approach	Characteristics
1	Traditional manual sorting	Low efficiency and health hazards for workers
2	Sensor-based hardware sorters (capacitive / inductive proximity)	Bulky equipment and high cost
3	Heavy CNNs (VGG16, ResNet50)	High accuracy, but 138M parameters make them too slow for web and mobile
4	MobileNetV2 transfer learning (proposed)	Lightweight with ~3.4M parameters, ideal for real-time edge and web use

MobileNetV2 offers the best balance of accuracy, size and speed for web deployment.

SYSTEM DESIGN

Proposed System Architecture



DATA

Dataset Specifications

Source: Kaggle, “Real-World Waste Classification Dataset - 5 Class”

Class	Images
Glass	314
Metal	375
Organic Waste	300
Paper	323
Plastic	335
Total	1,647

Zero data leakage split

70% **Training**
1,151 images

15% **Validation**
246 images

15% **Test**
250 images

Test images stay unseen during training and validation.

MODEL

Methodology and AI Model

MODEL STACK

1. MobileNetV2 base (ImageNet weights)
2. GlobalAveragePooling2D
3. BatchNormalization
4. Dense(128, ReLU)
5. Dropout(0.35)
6. Dense(5, Softmax)

TRAINING SETUP

Component	Configuration
Backbone	MobileNetV2
Core blocks	Depthwise separable convolutions and inverted residuals
Transfer learning	Pre-trained ImageNet weights
Loss	Categorical crossentropy
Optimizer	Adam (lr = 0.001)
Callbacks	EarlyStopping and ReduceLROnPlateau

DEPLOYMENT

Full-Stack Implementation

Flask backend

Flask Python backend loads the trained model and returns predictions.

Web frontend

HTML5, CSS3 and JavaScript interface for upload, live preview and results.

Live camera streaming

WebRTC MediaDevices API streams the webcam through navigator.mediaDevices.getUserMedia.

Asynchronous requests

Images travel as AJAX FormData, so predictions arrive without a page reload.

PREPROCESSING FORMULA

```
Normalized = (Pixel / 127.5) - 1.0
```

Maps pixel values from [0, 255] to [-1, 1] for MobileNetV2.

EVALUATION

Results and Evaluation

TEST ACCURACY

86.80%

on 250 unseen images

TEST LOSS

0.3646

categorical crossentropy

Class	Precision	Recall	F1-score
Glass	93.48%	89.58%	91.49%
Metal	82.26%	89.47%	85.71%
Organic Waste	97.56%	88.89%	93.02%
Paper	86.96%	81.63%	84.21%
Plastic	78.18%	84.31%	81.13%

Best F1: Organic Waste (93.02%). Lowest F1: Plastic (81.13%). The confusion matrix and training curves complete the evaluation.

WRAP-UP

Conclusion and Future Scope

SUMMARY

86.80%

Successfully demonstrated lightweight, high-accuracy waste classification with a live camera feed.

FUTURE SCOPE

1. **Multiple waste object detection**
   Detect several items in one frame with YOLOv8.

2. **IoT-enabled automated bins**
   Arduino or Raspberry Pi servo lids open for the predicted class.

3. **Cross-platform mobile app**
   Bring the classifier to everyday mobile devices.

SOURCES AND DEMO

References and Live System Demo

REFERENCES

1. M. Sandler, A. Howard, M. Zhu, A. Zhmoginov and L.-C. Chen, “MobileNetV2: Inverted Residuals and Linear Bottlenecks,” IEEE CVPR, 2018.
2. Kaggle, “Real-World Waste Classification Dataset - 5 Class.”
3. Yang & Thung (2016): Classification of Trash for Recyclability Status, Stanford CS229.

LIVE SYSTEM DEMO

Try it live

`http://127.0.0.1:5000`

- Quick sample image tests
- Live webcam prediction